Cloud Computing Software Development SE 4220

Project: Google Cloud

Creating an SQL-based Application using GCP

This project requires teams to develop a cloud-based photo gallery application leveraging Google Cloud Platform (GCP) services.

Create a cloud-based photo gallery application using the following architecture components:

1- Google Cloud SQL: Use Google Cloud SQL or MySql to create a relational database instance. This will store and manage the application's data.

2- Web Application Framework:

- Options: Flask (Python) or Node.js.
- Objective: Build the web interface of your photo gallery application using either Flask or Node.js.
- Recommendation: While both options are suitable, we encourage you to explore Node.js for its scalability and compatibility with cloud services.

3- Google Compute Engine (GCE):

- Objective: Deploy your application on Google Compute Engine to leverage scalable and secure virtual machines.
- Note: The choice of GCE emphasizes the use of Google Cloud services and provides an alternative to deploying on AWS EC2 instances.

Hints:

For a smooth start and efficient progress, refer to the "Account_Setup_GCP.pdf" guide uploaded in the modules section on Canvas under "Projects and labs. This document is crucial for setting up your GCP account and understanding the platform's services. Also:

- Consult Google Cloud SQL and Compute Engine documentation for technical guidance.
- Use Flask or Node.js documentation for web app development insights.

Deliverables:

Project Report (PDF format): Submit a comprehensive report including:

1. Screenshots demonstrating key functionalities:

    a) User login screen.
    b) Uploading of 10 photos.
    c) Searching and downloading of at least 2 photos.

2. Contribution Matrix with a brief description about each member role and contribution in the project.

3. Application Access:

    a) Provide a link to the deployed application.
    b) Include any necessary passwords, usernames for access (Ensure the application link and password are clearly indicated in your report).
    c) A new-user login to allow a new user to create an account and upload/download their own photos.

Evaluation Criteria: Your project will be evaluated based on the following criteria:

- Functionality (50%): The application meets all specified requirements.
- Usability (20%): The application is user-friendly and intuitive.
- Integration (20%): Effective use of GCP services as specified in the project components.
- Presentation (10%): Clarity and thoroughness of the project report.

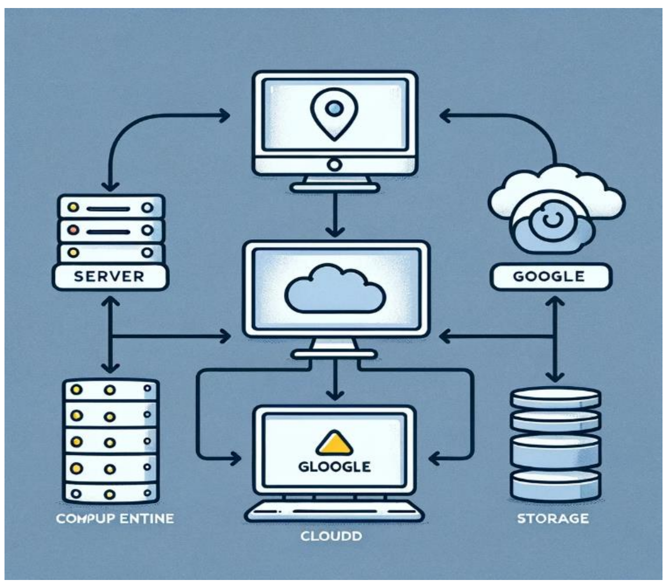

Figure 1: Project Architecture

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